

Electromagnetic Fields

Previous Year Questions • 1991–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Electromagnetic Fields (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 45 — GATE (IN)
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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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UNIT

I

Electrostatics

Topics

1. Coulomb's Law and Superposition Principle
2. Electric Field, Electric Field Intensity calculation in various Configurations: Due to Point, Line, Plane and Spherical Charge Distributions
3. Equipotential Surfaces and Electric Dipoles
4. Electric Flux Density and Gauss's Law
5. Divergence and Electric Potential
6. Electric Field Effect of Dielectric Medium- a)Boundary conditions b)method of image charges
7. Capacitance of Simple Configurations

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Unit Snapshot*

Stream: EE	**Questions:** 65	**MCQ:** 48	**MSQ:** 2	**NAT:** 15
1M: 32	**2M:** 28	**Descriptive:** 5	**Avg Weightage** ~3M	**Range:** 1M(2010)–6M(2020)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT 2026 2M

A uniform spherical volume charge distribution of radius 2 m, centered at the origin, has a strength of $\frac{3}{\pi}\times10^{-6}$ C/m^3. A point charge of strength $\pi \times 8.854 \times 10^{-12}$ C is moved from $(-3, 0, -4)$ to $(0, 0, 4)$ in Cartesian coordinate system. The relative permittivity of the medium is 1 and the coordinate values are in meters. The work done during the process is _____ μJ. (Round off to two decimal places)

Q1.26.2 MCQ 2026 1M

A uniform ring charge of radius R carries a total charge Q. Which one of the following options correctly quantifies the magnitude of the force on a point charge of strength q kept at the center of the ring?
(ϵ is the permittivity of the medium)

A. $\frac{Qq}{4\pi\epsilon R}$ **B.** $\frac{Qq}{4\pi\epsilon R^2}$

C. 0 **D.** $\frac{q}{4\pi\epsilon R}\times\frac{Q}{2\pi R}$

Q1.26.3 MCQ 2026 1M

A positive point charge with velocity $\vec{v} = 5\,\hat{x}$ enters a region having electric field $\vec{E} = 4\,\hat{y}$ and magnetic field $\vec{B} = -6\,\hat{z}$. Which one of the following statements is correct for the force on the charge as it enters the region?

A. The force will be along the magnetic field but perpendicular to the electric field

B. The force will be along the electric field but perpendicular to the magnetic field

C. The force will be perpendicular to both electric and magnetic field

D. The magnetic field does not exert any force on the charge

Q1.25.1 MCQ 2025 2M

An air filled cylindrical capacitor (capacitance C_0) of length L, with a and b as its inner and outer radii, respectively, consists of two coaxial conducting surfaces. Its cross-sectional view is shown in Fig. (i). In order to increase the capacitance, a dielectric material of relative permittivity ε_r is inserted inside 50% of the annular region as shown in Fig. (ii). The value of ε_r for which the capacitance of the capacitor in Fig. (ii), becomes $5C_0$ is

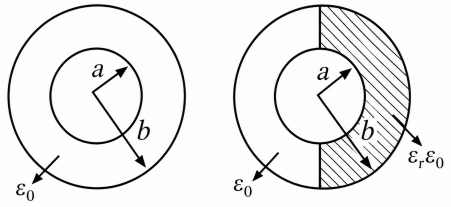


Fig. (i) Fig. (ii)

Fig. Q1.25.1

A. 4 **B.** 5 **C.** 9 **D.** 10

Q2.25.2 MSQ 2025 2M

Let a_R be the unit radial vector in the spherical co-ordinate system. For which of the following value(s) of n, the divergence of the radial vector field $f(R) = a_R\frac{1}{R^n}$ is independent of R?

A. -2 **B.** -1 **C.** 1 **D.** 2

Q1.25.3 MCQ 2025 1M

Which one of the following figures represents the radial electric field distribution E_R caused by a spherical cloud of electrons with a volume charge density, $\rho = -3\rho_0$ for $0 \le R \le a$ (both ρ_0, a are positive and R is the radial distance) and $\rho = 0$ for $R > a$?

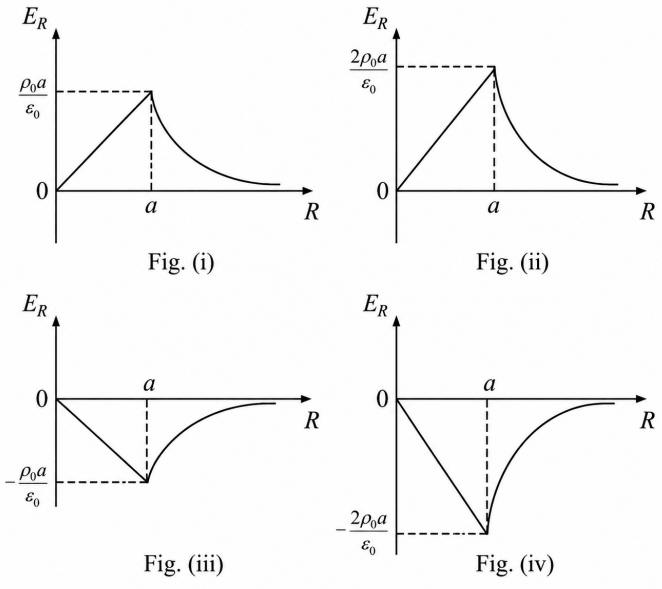


Fig. (i) Fig. (ii) Fig. (iii) Fig. (iv)

Fig. Q1.25.3

A. Fig. (i) **B.** Fig. (ii) **C.** Fig. (iii) **D.** Fig. (iv)

Q1.24.1 NAT 2024 2M

In the (x, y, z) coordinate system, three point-charges Q, Q and αQ are located in free space at $(-1, 0, 0)$, $(1, 0, 0)$ and $(0, -1, 0)$ respectively. The value of α for the electric field to be zero at $(0, 0.5, 0)$ is _____ (rounded off to 1 decimal places).

Q1.22.1 MCQ 2022 2M

As shown in the figure below, two concentric conducting spherical shells, centered at $r = 0$ and having radii $r = c$ and $r = d$ are maintained at potentials such that the potential $V(r)$ at $r = c$ is V_1 and $V(r)$ at $r = d$ is V_2. Assume that $V(r)$ depends only on r, where r is the radial distance. The expression for $V(r)$ in the region between $r = c$ and $r = d$ is

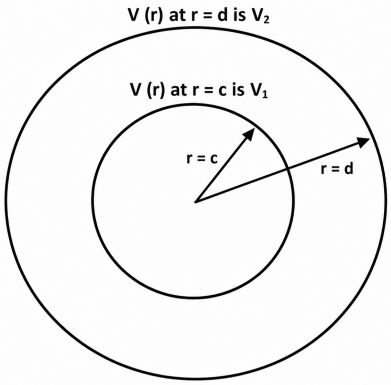


Fig. Q1.22.1

A. $V(r) = \dfrac{cd(V_2 - V_1)}{(d-c)r} - \dfrac{V_1 c + V_2 d - 2V_1 d}{d-c}$

B. $V(r) = \dfrac{cd(V_1 - V_2)}{(d-c)r} + \dfrac{V_2 d - V_1 c}{d-c}$

C. $V(r) = \dfrac{cd(V_1 - V_2)}{(d-c)r} - \dfrac{V_1 c - V_2 c}{d-c}$

D. $V(r) = \dfrac{cd(V_2 - V_1)}{(d-c)r} - \dfrac{V_2 c - V_1 c}{d-c}$

Q1.21.1 NAT 2021 1M

A 1 μC point charge is held at the origin of a cartesian coordinate system. If a second point charge of 10 μC is moved from $(0, 10, 0)$ to $(5, 5, 5)$ and subsequently to $(5, 0, 0)$, then the total work done is _____ mJ. (Round off to 2 decimal places).

Take $\dfrac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ in SI units. All coordinates are in meters.

Q1.21.2 NAT 2021 2M

Consider a large parallel plate capacitor. The gap 'd' between the two plates is filled entirely with a dielectric slab of relative permittivity 5. The plates are initially charged to a potential difference of V volts and then disconnected from the source. If the dielectric slab is pulled out completely, then the ratio of the new electric field E_2 in the gap to the original electric field E_1 is _____.

Q1.20.1 MCQ 2020 2M

Let a_r, a_ϕ and a_z be unit vectors along r, ϕ and z directions, respectively in the cylindrical coordinate system. For the electric flux density given by $D = (a_r 15 + a_\phi 2r - a_z 3rz)$ Coulomb/m^2, the total electric flux, in Coulomb, emanating from the volume enclosed by a solid cylinder of radius 3 m and height 5 m oriented along the z-axis with its base at the origin is:

A. 108π **B.** 54π **C.** 90π **D.** 180π

Q1.20.2 MCQ 2020 2M

The static electric field inside a dielectric medium with relative permittivity, $\varepsilon_r = 2.25$, expressed in cylindrical coordinate system is given by the following expression

$$E = a_r 2r + a_\varphi \left(\frac{3}{r}\right) + a_z 6$$

where a_r, a_φ, a_z are unit vectors along r, φ and z directions, respectively. If the above expression represents a valid electrostatic field inside the medium, then the volume charge density associated with this field in terms of free space permittivity, ε_0, in SI units is given by:

A. $3\varepsilon_0$ **B.** $4\varepsilon_0$ **C.** $5\varepsilon_0$ **D.** $9\varepsilon_0$

Q1.20.3 MCQ 2020 2M

The vector function expressed by $\mathbf{F} = a_x(5y - k_1 z) + a_y(3z + k_2 x) + a_z(k_3 y - 4x)$ represents a conservative field, where a_x, a_y, a_z are unit vectors along x, y and z directions, respectively. The values of constants k_1, k_2, k_3 are given by:

A. $k_1 = 3,\ k_2 = 3,\ k_3 = 7$ **B.** $k_1 = 3,\ k_2 = 8,\ k_3 = 5$
C. $k_1 = 4,\ k_2 = 5,\ k_3 = 3$ **D.** $k_1 = 0,\ k_2 = 0,\ k_3 = 0$

Q1.18.1 NAT 2018 2M

The capacitance of an air-filled parallel-plate capacitor is 60 pF. When a dielectric slab whose thickness is half the distance between the plates, is placed on one of the plates covering it entirely, the capacitance becomes 86 pF. Neglecting the fringing effects, the relative permittivity of the dielectric is _____ (up to 2 decimal places).

Q1.18.2 MCQ 2018 1M

A positive charge of 1 nC is placed at $(0, 0, 0.2)$ where all dimensions are in metres. Consider the x-y plane to be a conducting ground plane. Take $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m. The Z component of the E field at $(0, 0, 0.1)$ is closest to

A. 899.18 V/m **B.** −899.18 V/m
C. 999.09 V/m **D.** −999.09 V/m

Q1.17.1 NAT 2017 2M

A thin soap bubble of radius $R = 1$ cm and thickness $a = 3.3$ μm ($a \ll R$) is at a potential of 1 V with respect to a reference point at infinity. The bubble bursts and becomes a single spherical drop of soap (assuming all the soap is contained in the drop) of radius r. The volume of the soap in the thin bubble is $4\pi R^2 a$ and that of the drop is $\frac{4}{3}\pi r^3$. The potential, in volts, of the resulting single spherical drop with respect to the same reference point at infinity is _____. (Give the answer up to two decimal places.)

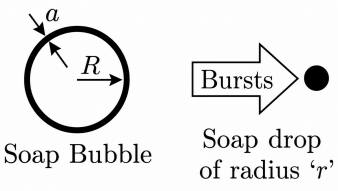


Fig. Q1.17.1

Q1.17.2 MCQ 2017 1M

Consider a solid sphere of radius 5 cm made of a perfect electric conductor. If one million electrons are added to this sphere, these electrons will be distributed

A. uniformly over the entire volume of the sphere

B. uniformly over the outer surface of the sphere

C. concentrated around the centre of the sphere

D. along a straight line passing through the centre of the sphere

Q1.17.3 MCQ 2017 1M

The figures show diagrammatic representations of vector fields $\vec{X}$, $\vec{Y}$ and $\vec{Z}$ respectively. Which one of the following choices is true?

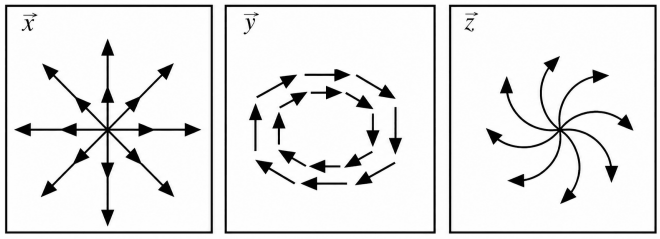


Fig. Q1.17.3

A. $\nabla \cdot \vec{X} = 0, \nabla \times \vec{Y} \neq 0, \nabla \times \vec{Z} = 0$

B. $\nabla \cdot \vec{X} \neq 0, \nabla \times \vec{Y} = 0, \nabla \times \vec{Z} \neq 0$

C. $\nabla \cdot \vec{X} \neq 0, \nabla \times \vec{Y} \neq 0, \nabla \times \vec{Z} \neq 0$

D. $\nabla \cdot \vec{X} = 0, \nabla \times \vec{Y} = 0, \nabla \times \vec{Z} = 0$

Q1.17.4 MCQ 2017 1M

Consider an electron, a neutron and a proton initially at rest and placed along a straight line such that the neutron is exactly at the center of the line joining the electron and proton. At $t = 0$, the particles are released but are constrained to move along the same straight line. Which of these will collide first?

A. The particles will never collide

B. All will collide together

C. Proton and Neutron

D. Electron and Neutron

Q1.16.1 MCQ 2016 1M

Two electrodes, whose cross-sectional view is shown in the figure below, are at the same potential. The maximum electric field will be at the point

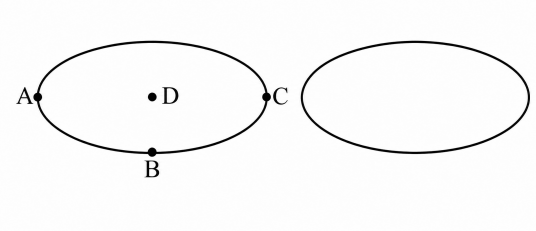


Fig. Q1.16.1

A. A **B.** B **C.** C **D.** D

Q1.16.2 MCQ 2016 1M

A parallel plate capacitor filled with two dielectrics is shown in the figure below. If the electric field in the region A is 4 kV/cm, the electric field in the region B, in kV/cm, is

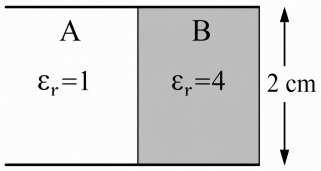


Fig. Q1.16.2

A. 1 **B.** 2 **C.** 4 **D.** 16

Q1.16.3 MCQ 2016 2M

Two electric charges q and $-2q$ are placed at $(0, 0)$ and $(6, 0)$ on the x-y plane. The equation of the zero equipotential curve in the x-y plane is

A. $x = -2$ **B.** $y = 2$
C. $x^2 + y^2 = 2$ **D.** $(x + 2)^2 + y^2 = 16$

Q1.16.4 MCQ 2016 1M

In cylindrical coordinate system, the potential produced by a uniform ring charge is given by $\varphi = f(r, z)$, where f is a continuous function of r and z. Let $\vec{E}$ be the resulting electric field. Then the magnitude of $\nabla \times \vec{E}$ is

A. increases with r **B.** is 0
C. is 3 **D.** decreases with z

Q1.15.1 MCQ 2015 1M

Two semi-infinite dielectric regions are separated by a plane boundary at $y = 0$. The dielectric constants of region 1 ($y < 0$) and region 2 ($y > 0$) are 2 and 5, respectively. Region 1 has uniform electric field $\vec{E} = 3\hat{a}_x + 4\hat{a}_y + 2\hat{a}_z$, where $\hat{a}_x$, $\hat{a}_y$, and $\hat{a}_z$ are unit vectors along the x, y and z axes, respectively. The electric field in region 2 is

A. $3\hat{a}_x + 1.6\hat{a}_y + 2\hat{a}_z$ **B.** $1.2\hat{a}_x + 4\hat{a}_y + 2\hat{a}_z$
C. $1.2\hat{a}_x + 4\hat{a}_y + 0.8\hat{a}_z$ **D.** $3\hat{a}_x + 10\hat{a}_y + 0.8\hat{a}_z$

Q1.15.2 MCQ 2015 2M

Two semi-infinite conducting sheets are placed at right angles to each other as shown in the figure. A point charge of $+Q$ is placed at a distance of d from both sheets. The net force on the charge is $\frac{Q^2}{4\pi\varepsilon_0}\frac{K}{d^2}$, where K is given by

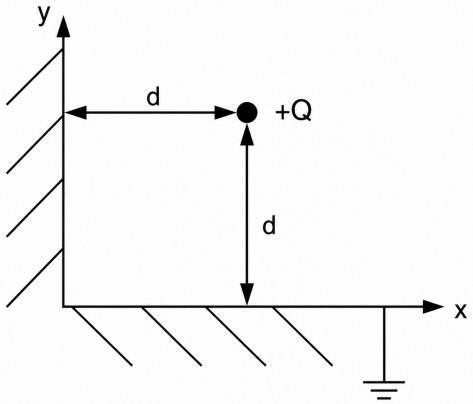


Fig. Q1.15.2

A. 0 **B.** $-\frac{1}{4}\hat{i} - \frac{1}{4}\hat{j}$
C. $-\frac{1}{8}\hat{i} - \frac{1}{8}\hat{j}$ **D.** $\frac{1-2\sqrt{2}}{8\sqrt{2}}\hat{i} + \frac{1-2\sqrt{2}}{8\sqrt{2}}\hat{j}$

Q1.15.3 NAT 2015 2M

A parallel plate capacitor is partially filled with glass of dielectric constant 4.0 as shown below. The dielectric strength of air and glass are 30 kV/cm and 300 kV/cm, respectively. The maximum voltage (in kilovolts) which can be applied across the capacitor without any breakdown is

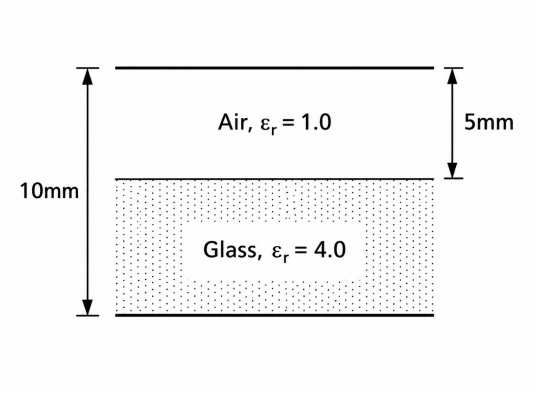


Fig. Q1.15.3

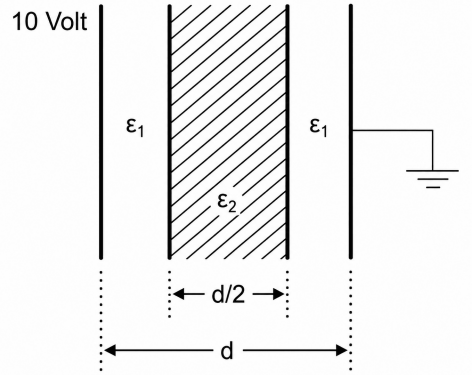


Fig. Q1.14.2

Q1.14.1 MCQ 2014 1M

C_0 is the capacitance of a parallel plate capacitor with air as dielectric (as in figure (a)). If, half of the entire gap as shown in figure (b) is filled with a dielectric of permittivity ε_r, the expression for the modified capacitance is

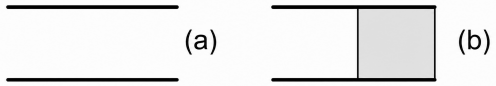


Fig. Q1.14.1

A. $\frac{C_0}{2}(1+\varepsilon_r)$ **B.** $(C_0+\varepsilon_r)$

C. $\frac{C_0}{2}\varepsilon_r$ **D.** $C_0(1+\varepsilon_r)$

Q1.14.2 MCQ 2014 1M

A parallel plate capacitor consisting two dielectric materials is shown in the figure. The middle dielectric slab is placed symmetrically with respect to the plates.

If the potential difference between one of the plates and the nearest surface of dielectric interface is 2 Volts, then the ratio $\varepsilon_1 : \varepsilon_2$ is

A. 1 : 4 **B.** 2 : 3 **C.** 3 : 2 **D.** 4 : 1

Q1.14.3 MCQ 2014 1M

Let $\nabla \cdot (f\,\mathbf{v}) = x^2y + y^2z + z^2x$, where f and $\mathbf{v}$ are scalar and vector fields respectively. If $\mathbf{v} = y\,\mathbf{i} + z\,\mathbf{j} + x\,\mathbf{k}$, then $\mathbf{v} \cdot \nabla f$ is

A. $x^2y + y^2z + z^2x$ **B.** $2xy + 2yz + 2zx$
C. $x + y + z$ **D.** 0

Q1.14.4 MCQ 2014 2M

A perfectly conducting metal plate is placed in x-y plane in a right handed coordinate system. A charge of $+32\pi\varepsilon_0\sqrt{2}$ columbs is placed at coordinate $(0, 0, 2)$. ε_0 is the permittivity of free space. Assume $\hat{i}$, $\hat{j}$, $\hat{k}$ to be unit vectors along x, y and z axes respectively. At the coordinate $(\sqrt{2}, \sqrt{2}, 0)$, the electric field vector $\vec{E}$ (Newtons/Columb) will be

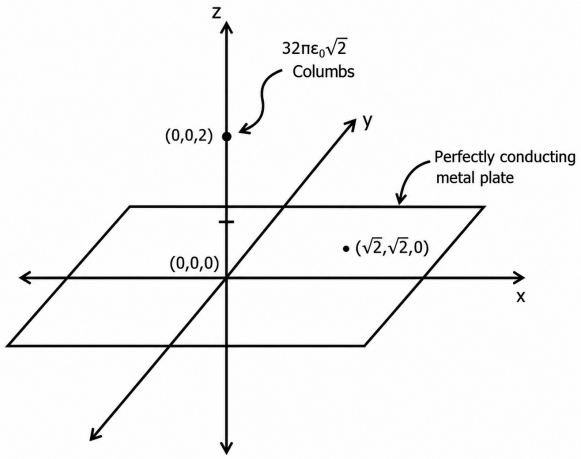


Fig. Q1.14.4

A. $2\sqrt{2}\,\hat{k}$ **B.** $-2\,\hat{k}$ **C.** $2\,\hat{k}$ **D.** $-2\sqrt{2}\,\hat{k}$

Q1.14.5 MCQ 2014 1M

A hollow metallic sphere of radius r is kept at potential of 1 Volt. The total electric flux coming out of the concentric spherical surface of radius $R(> r)$ is

A. $4\pi\varepsilon_0 r$ **B.** $4\pi\varepsilon_0 r^2$ **C.** $4\pi\varepsilon_0 R$ **D.** $4\pi\varepsilon_0 R^2$

Q1.13.1 MCQ 2013 1M

The curl of the gradient of the scalar field defined by $V = 2x^2y + 3y^2z + 4z^2x$ is

A. $4xy\,a_x + 6yz\,a_y + 8zx\,a_z$

B. $4a_x + 6a_y + 8a_z$

C. $(4xy + 4z^2)a_x + (2x^2 + 6yz)a_y + (3y^2 + 8zx)a_z$

D. 0

Q1.13.2 MCQ 2013 1M

Given a vector field $\mathbf{F} = y^2x\,a_x - yz\,a_y - x^2\,a_z$, the line integral $\int \mathbf{F} \cdot d\mathbf{l}$ evaluated along a segment on the x-axis from $x = 1$ to $x = 2$ is

A. -2.33 **B.** 0 **C.** 2.33 **D.** 7

Q1.13.3 MCQ 2013 2M

Three capacitors C_1, C_2, and C_3, whose values are 10 μF, 5 μF, and 2 μF respectively, have breakdown voltages of 10 V, 5 V, and 2 V respectively. For the interconnection shown, the maximum safe voltage in Volts that can be applied across the combination and the corresponding total charge in μC stored in the effective capacitance across the terminals are respectively

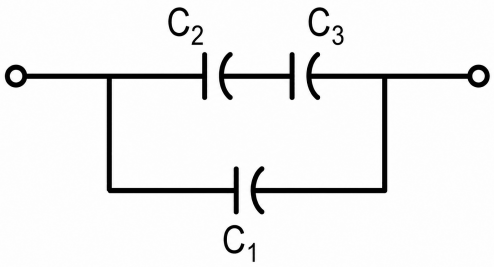


Fig. Q1.13.3

A. 2.8 and 36 **B.** 7 and 119
C. 2.8 and 32 **D.** 7 and 80

Q1.13.4 MCQ 2013 2M

A dielectric slab with 500 mm × 500 mm cross-section is 0.4 m long. The slab is subjected to a uniform electric field of $\mathbf{E} = 6a_x + 8a_y$ kV/mm. The relative permittivity of the dielectric material is equal to 2. The value of constant ε_0 is 8.85×10^{-12} F/m. The energy stored in the dielectric in Joules is

A. 8.85×10^{-11} **B.** 8.85×10^{-5} **C.** 88.5 **D.** 885

Q1.11.1 MCQ 2011 2M

A capacitor is made with a polymeric dielectric having an ε_r of 2.26 and a dielectric breakdown strength of 50 kV/cm. The permittivity of free space is 8.85 pF/m. If the rectangular plates of the capacitor have a width of 20 cm and a length of 40 cm, then the maximum electric charge in the capacitor is

A. 2 μC **B.** 4 μC **C.** 8 μC **D.** 10 μC

Q1.10.1 MCQ 2010 1M

Divergence of the three-dimensional radial vector field $\vec{r}$ is

A. 3 **B.** $1/r$ **C.** $\hat{i} + \hat{j} + \hat{k}$ **D.** $3(\hat{i} + \hat{j} + \hat{k})$

Q1.08.1 MCQ 2008 2M

A capacitor consists of two metal plates each 500×500 mm^2 and spaced 6 mm apart. The space between the metal plates is filled with a glass plate of 4 mm thickness and a layer of paper of 2 mm thickness. The relative permittivities of the glass and paper are 8 and 2 respectively. Neglecting the fringing effect, the capacitance will be

(Given that $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m)

A. 983.33 pF **B.** 1475 pF
C. 6637.5 pF **D.** 9956.25 pF

Q1.08.2 MCQ 2008 2M

Two point charges $Q_1 = 10\,\mu$C and $Q_2 = 20\,\mu$C are placed at coordinates $(1, 1, 0)$ and $(-1, -1, 0)$ respectively. The total electric flux passing through a plane $z = 20$ will be

A. 7.5 μC **B.** 13.5 μC **C.** 15.0 μC **D.** 22.5 μC

Q1.07.1 MCQ 2007 2M

A solid sphere made of insulating material has a radius R and has a total charge Q distributed uniformly in its volume. What is the magnitude of the electric field intensity, E, at a distance r $(0 < r < R)$ inside the sphere?

A. $\frac{1}{4\pi\varepsilon_0}\frac{Qr}{R^3}$ **B.** $\frac{3}{4\pi\varepsilon_0}\frac{Qr}{R^3}$
C. $\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$ **D.** $\frac{1}{4\pi\varepsilon_0}\frac{QR}{r^3}$

Q1.07.2 MCQ 2007 1M

Divergence of the vector field $\mathbf{V}(x, y, z) = -(x\cos xy + y)\mathbf{i} + (y\cos xy)\mathbf{j} + \left(\sin z^2 + x^2 + y^2\right)\mathbf{k}$ is

A. $2z\cos z^2$ **B.** $\sin xy + 2z\cos z^2$
C. $x\sin xy - \cos z$ **D.** none of these

Q1.05.1 MCQ 2005 1M

If $\vec{E}$ is the electric field intensity, $\nabla \cdot (\nabla \times \vec{E})$ is equal to

A. $\vec{E}$ **B.** $|\vec{E}|$ **C.** null vector **D.** zero

Q1.04.1 MCQ 2004 1M

A parallel plate capacitor is shown in figure Q.2. It is made of two square metal plates of 400 mm side. The 14 mm space between the plates is filled with two layers of dielectrics of $\varepsilon_r = 4$, 6 mm thick and $\varepsilon_r = 2$, 8 mm thick. Neglecting fringing of fields at the edges the capacitance is

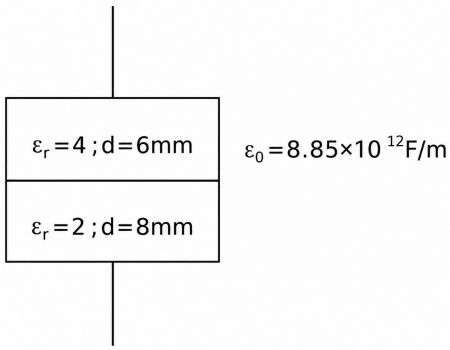


Fig. Q1.04.1

A. 1298 pF **B.** 944 pF **C.** 354 pF **D.** 257 pF

Q1.03.1 MCQ 2003 2M

A point charge of +1 nC is placed in a space with a permitivity of 8.85×10^{-12} F/m as shown in Fig.Q38. The potential difference V_{PQ} between two points P and Q at distances of 40 mm and 20 mm respectively from the point charge is

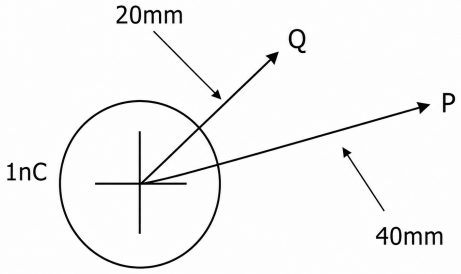


Fig. Q1.03.1

A. 0.22 kV **B.** −225 V **C.** −2.24 kV **D.** 15 V

Q1.03.2 MCQ 2003 2M

A parallel plate capacitor has an electrode area of 100 mm^2, with a spacing of 0.1 mm between the electrodes. The dielectric between the plates is air with a permittivity of 8.85×10^{-12} F/m. The charge on the capacitor is 100 V. the stored energy in the capacitor is

A. 8.85 pJ **B.** 440 pJ **C.** 22.1 nJ **D.** 44.3 nJ

Q1.03.3 MCQ 2003 2M

A composite parallel plate capacitor is made up of two different dielectric materials with different thickness (t_1 and t_2) as shown in Fig.Q40. The two different dielectric materials are separated by a conducting foil F. The voltage of the conducting foil is

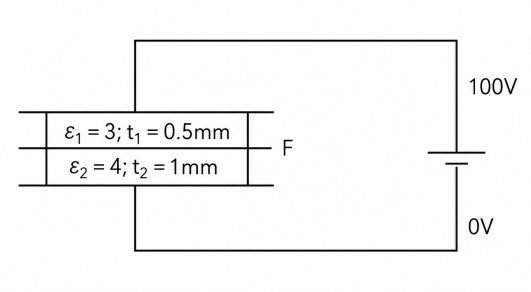


Fig. Q1.03.3

A. 52 V **B.** 60 V **C.** 67 V **D.** 33 V

Q1.01.1 MCQ 2001 2M

The electric field $\vec{E}$ (in volts/metre) at the point (1, 1, 0) due to a point charge of $+1\mu$C located at $(-1, 1, 1)$ (coordinates in metres) is

A. $\frac{10^{-6}}{20\sqrt{5}\pi\varepsilon_0}(2\vec{i} - \vec{k})$ **B.** $\frac{10^{-6}}{20\pi\varepsilon_0}(2\vec{i} - \vec{k})$

C. $\frac{-10^{-6}}{20\sqrt{5}\pi\varepsilon_0}(2\vec{i} - \vec{k})$ **D.** $\frac{-10^{-6}}{20\pi\varepsilon_0}(2\vec{i} - \vec{k})$

Q1.01.2 MCQ 2001 2M

Given the potential function in free space to be $V(x) = (50x^2 - 50y^2 + 50z^2)$ volts, the magnitude (in volts/metre) and the direction of the electric field at a point $(1, -1, 1)$, where the dimensions are in metres, are

A. 100; $(\vec{i} + \vec{j} + \vec{k})$ **B.** $\frac{100}{\sqrt{3}}$; $(\vec{i} - \vec{j} + \vec{k})$

C. $\frac{100}{\sqrt{3}}$; $\left[\frac{(-\vec{i} - \vec{j} - \vec{k})}{\sqrt{3}}\right]$ **D.** $\frac{100}{\sqrt{3}}$; $\left[\frac{(\vec{i} - \vec{j} - \vec{k})}{\sqrt{3}}\right]$

Q1.99.1 NAT 1999 5M

A charge $+Q$ is uniformly distributed throughout the volume of a dielectric sphere of radius R and dielectric constant ε_r, based on Gauss law, determine the expressions for the electric field E as a function of distance r from the center of the sphere, within the ranges $0 < r < R$ and $R \leq r$. Indicate expression(s) for the critical point(s) on the sketch.

Q1.99.2 MCQ 1999 1M

When the plate area of a parallel plate capacitor is increased keeping the capacitor voltage constant, the force between the plates

A. increase

B. decreases

C. remains constant

D. may increase or decrease depending on the metal making up the plates

Q1.98.1 NAT 1998 5M

An infinite number of charges, each equal to q are placed along the $x = 1$, $x = 2$, $x = 4$, $x = 8$, $x = 16$ and so on. Find the potential and electron field at point $x = 0$, due to these system of charges.

Q1.98.2 MCQ 1998 1M

The laws of electromagnetic induction (Faraday's and Lenz's law) are summarized in the following equation:

A. $e = iR$ **B.** $e = L\frac{dI}{dt}$

C. $e = -\frac{d\psi}{dt}$ **D.** None of the above

Q1.97.1 MCQ 1997 1M

In a uniform electric field, field lines and equipotentials

A. Are parallel to one another **B.** Intersect at 45°

C. Intersect at 30° **D.** Are orthogonal

Q1.97.2 NAT 1997 2M

The capacitance of the arrangement shown in figure is _____ pF

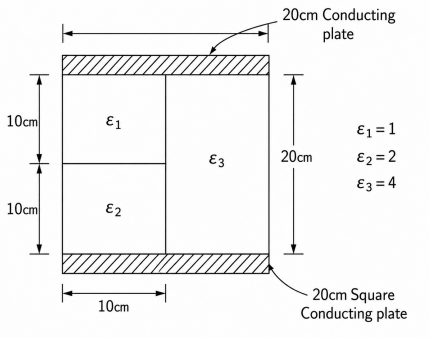


Fig. Q1.97.2

Q1.97.3 NAT 1997 2M

The capacitance of an isolated sphere of radius 10cm in air is equal to _____ pF

Q1.97.4 NAT 1997 5M

Determine the electric field intensity at the point p for the arrangement shown in figure above

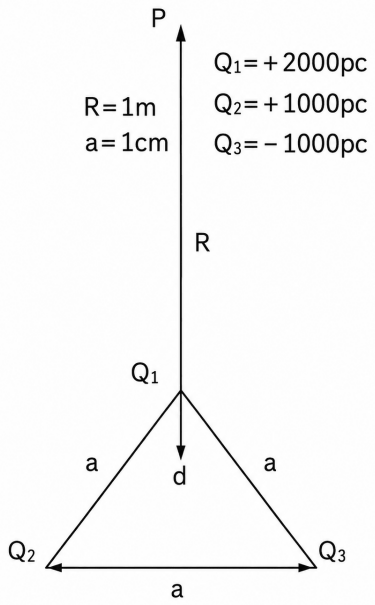


Fig. Q1.97.4

Q1.96.1 MCQ 1996 1M

If v, w, q stand for voltage, energy and charge, then v can be expressed as

A. $v = \frac{dq}{dw}$ **B.** $v = \frac{dw}{dq}$

C. $dv = \frac{dw}{dq}$ **D.** $dv = \frac{dq}{dw}$

Q1.96.2 MCQ 1996 1M

Inside a hollow conducting sphere

A. Electric field is zero

B. Electric field is an non-zero constant

C. Electric field changes with the magnitude of the charge given to the conductor

D. Electric field changes with distance from the centre the sphere

Q1.95.1 MCQ 1995 1M

A spherical conductor of radius 'a' with charge 'q' is placed concentrically inside an uncharged and unearthed spherical conducting shell of inner and outer radii r_1 and r_2 respectively. Taking potential to be zero at infinity, the potential at any point P within the shell ($r_1 < r < r_2$) will be

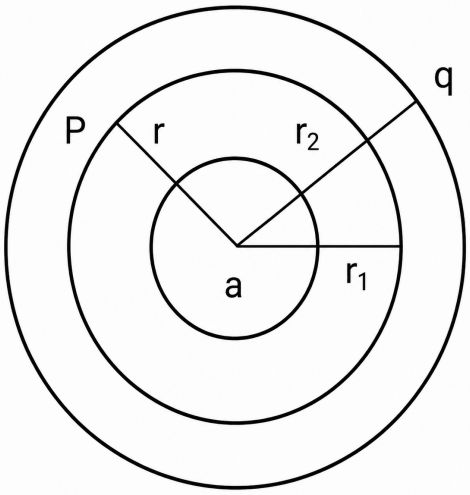


Fig. Q1.95.1

A. $\frac{q}{4\pi\varepsilon_0 r}$ **B.** $\frac{q}{4\pi\varepsilon_0 a}$

C. $\frac{q}{4\pi\varepsilon_0 r_2}$ **D.** $\frac{q}{4\pi\varepsilon_0 r_1}$

Q1.94.1 NAT 1994 1M

A hollow conductor is at a potential V. The potential at any point inside the hollow is _____

Q1.94.2 NAT 1994 1M

In electrostatic field $\nabla \times \vec{E} = 0$

Q1.94.3 MCQ 1994 1M

When a charge is given to a conductor

A. It distributes uniformly all over the surface

B. It distributes uniformly all over the volume

C. It distributes on the surface, inversely proportional to the radius of curvature

D. It stays where it was placed

Q1.92.1 NAT 1992 1M

An electrostatic potential is given by $\phi = 2x\sqrt{y}$ volts in the rectangular co-ordinate system. The magnitude of the electric field at $x = 1$ m, $y = 1$ m is _____ V/m (rounded off to three decimal places).

Q1.92.2 MSQ 1992 1M

Which of the following equations represents the Gauss' law in a homogeneous isotropic medium?

A. $\oiint \vec{D} \cdot d\vec{s} = \iiint \rho \, dv$

B. $\nabla \times \vec{H} = \vec{D}$

C. $\nabla \cdot \vec{J} + \rho = 0$

D. $\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon}$

Q1.91.1 MCQ 1991 1M

The mica layer ($\varepsilon_r = 7$) in a parallel plate capacitor with an effective area of 120 mm^2 has a damaged section equivalent to a hole of 0.5 mm diameter. Which of the following would be significantly affected by the damage?

A. Capacitance

B. Charge

C. Dielectric breakdown strength

D. $\tan\delta$

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q1.26.1 0.38 to 0.42	Q1.26.2 C	Q1.26.3 B
Q1.25.1 C	Q2.25.2 B,D	Q1.25.3 C
Q1.24.1 -1.7 to -1.5	Q1.22.1 B	Q1.21.1 8.90 to 9.10
Q1.21.2 5 to 5	Q1.20.1 D	Q1.20.2 D
Q1.20.3 C	Q1.18.1 2.50 to 2.55	Q1.18.2 D
Q1.17.1 9.5 to 10.5	Q1.17.2 B	Q1.17.3 C
Q1.17.4 D	Q1.16.1 A	Q1.16.2 C
Q1.16.3 D	Q1.16.4 B	Q1.15.1 A
Q1.15.2 D	Q1.15.3 18.75	Q1.14.1 A
Q1.14.2 C	Q1.14.3 A	Q1.14.4 B
Q1.14.5 A	Q1.13.1 D	Q1.13.2 B
Q1.13.3 C	Q1.13.4 C	Q1.11.1 C
Q1.10.1 A	Q1.08.1 B	Q1.08.2 C
Q1.07.1 A	Q1.07.2 A	Q1.05.1 D
Q1.04.1 D	Q1.03.1 B	Q1.03.2 D
Q1.03.3 B	Q1.01.1 A	Q1.01.2 *
Q1.99.1 *	Q1.99.2 A	Q1.98.1 *
Q1.98.2 C	Q1.97.1 D	Q1.97.2 4.72
Q1.97.3 11.1	Q1.97.4 *	Q1.96.1 C
Q1.96.2 A	Q1.95.1 D	Q1.94.1 V
Q1.94.2 True	Q1.94.3 C	Q1.92.1 2.236
Q1.92.2 A,D	Q1.91.1 C	

UNIT

II

Magnetostatics

Topics

1. Biot-Savart's Law
2. Electromagnetic Induction and Faraday's Law
3. Magnetic Field of a current carrying loop
4. Ampere's Law and Curl
5. Lorentz Force
6. Inductance and Magnetomotive Force
7. Self and Mutual Inductance of Simple Configurations
8. Reluctance and Magnetic Circuits
9. Magnetic Flux and Boundary Conditions for magnetic field

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 43	**MCQ:** 24	**MSQ:** 1	**NAT:** 18
1M: 25	**2M:** 11	**Descriptive:** 7	**Avg Weightage** ~2M	**Range:** 1M(2026)–4M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 1M

The figure shows an arbitrarily shaped planar conducting loop A in the XY plane. Two nonintersecting regions with areas a_1 and a_2 within the loop are subjected to magnetic fields $\vec{B}_1 = \frac{m}{\sqrt{2}} \sin(\omega t)\,(1\,\hat{x} + 0\,\hat{y} + 1\,\hat{z})$ and $\vec{B}_2 = -\frac{n}{\sqrt{2}} \cos\left(2\omega t + \frac{\pi}{4}\right)(0\,\hat{x} + 1\,\hat{y} + 1\,\hat{z})$, respectively.

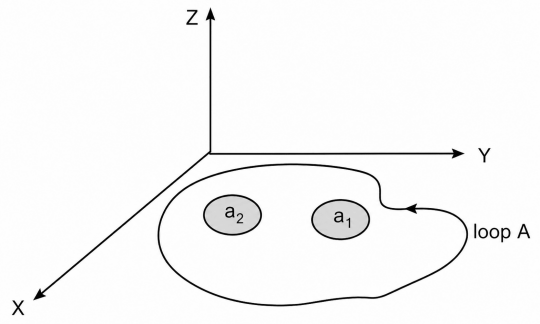


Fig. Q2.26.1

What is the expression for the induced RMS voltage in loop A?

A. $\sqrt{\frac{a_1^2\omega^2m^2 + 4a_2^2\omega^2n^2}{4}}$ **B.** $\sqrt{\frac{a_1^2\omega^2m^2 + 4a_2^2\omega^2n^2}{2}}$

C. $\sqrt{\frac{a_1^2\omega^2m^2 - 2a_2^2\omega^2n^2}{2}}$ **D.** $\sqrt{a_1^2\omega^2m^2 + 2a_2^2\omega^2n^2}$

Q2.24.1 NAT 2024 2M

The given equation represents a magnetic field strength $\vec{H}(r, \theta, \phi)$ in the spherical coordinate system, in free space. Here, $\hat{r}$ and $\hat{\theta}$ represent the unit vectors along r and θ, respectively. The value of P in the equation should be rounded off to the nearest integer.

$$\vec{H}(r, \theta, \phi) = \frac{1}{r^3}\left(P\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta}\right)$$

Q2.23.1 MSQ 2023 2M

In the figure, the electric field E and the magnetic field B point to x and z directions, respectively, and have constant magnitudes. A positive charge 'q' is released from rest at the origin. Which of the following statement(s) is/are true?

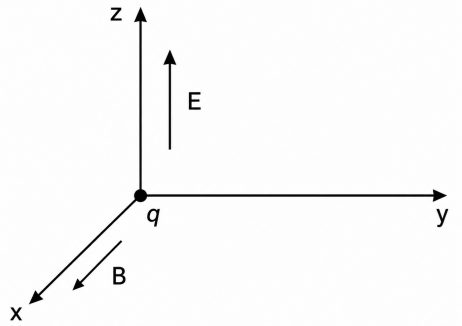


Fig. Q2.23.1

A. The charge will move in the direction of z with constant velocity.

B. The charge will always move on the y-z plane only.

C. The trajectory of the charge will be a circle.

D. The charge will progress in the direction of y.

Q2.23.2 NAT 2023 2M

An infinite surface of linear current density $\mathbf{K} = 5\hat{a}_x$ A/m exists on the x-y plane, as shown in the figure. The magnitude of the magnetic field intensity (H) at a point $(1, 1, 1)$ due to the surface current in Ampere/meter is _____ (Round off to 2 decimal places).

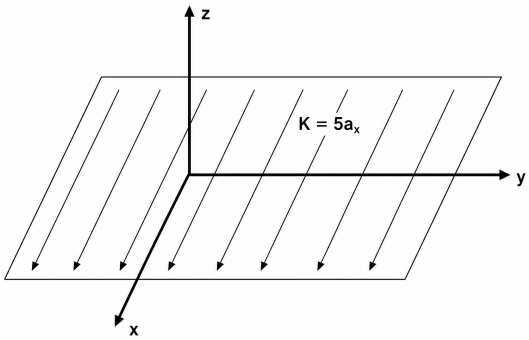


Fig. Q2.23.2

Q2.22.1 MCQ 2022 1M

A long conducting cylinder having a radius b is placed along the z axis. The current density is $\vec{J} = J_a r^3\,\hat{z}$ for the region $r < b$ where r is the distance in the radial direction. The magnetic field intensity (H) for the region inside the conductor (i.e. for $r < b$) is

A. $\frac{J_a}{4} r^4$ **B.** $\frac{J_a}{3} r^3$ **C.** $\frac{J_a}{5} r^4$ **D.** $J_a r^3$

Q2.22.2 MCQ 2022 2M

If the magnetic field intensity (H) in a conducting region is given by the expression $\mathbf{H} = x^2\hat{i} + x^2y^2\hat{j} + x^2y^2z^2\hat{k}$ A/m. The magnitude of the current density, in A/m^2, at $x = 1$ m, $y = 2$ m, and $z = 1$ m, is

A. 8 **B.** 12 **C.** 16 **D.** 20

Q2.21.1 MCQ 2021 1M

Which one of the following vector functions represents a magnetic field $\vec{B}$?

Here, $\hat{x}$, $\hat{y}$ and $\hat{z}$ are unit vectors along the x-, y- and z-axes, respectively.

A. $10x\hat{x} + 20y\hat{y} - 30z\hat{z}$

B. $10y\hat{x} + 20x\hat{y} - 10z\hat{z}$

C. $10z\hat{x} + 20y\hat{y} - 30x\hat{z}$

D. $10x\hat{x} - 30z\hat{y} + 20y\hat{z}$

Q2.21.2 NAT 2021 2M

One coulomb of point charge moving with a uniform velocity $10\,\hat{x}$ m/s enters the region $x \geq 0$ having a magnetic flux density $\vec{B} = (10y\,\hat{x} + 10x\,\hat{y} + 10\,\hat{z})$ T.

The magnitude of force on the charge at $x = 0^+$ is _____ N.

Here, $\hat{x}$, $\hat{y}$ and $\hat{z}$ are unit vectors along the x-, y- and z-axes, respectively.

Q2.20.1 NAT 2020 2M

A conducting square loop of side length 1 m is placed at a distance of 1 m from a long straight wire carrying a current $I = 2$ A as shown below. The mutual inductance, in nH (rounded off to 2 decimal places), between the conducting loop and the long wire is _____.

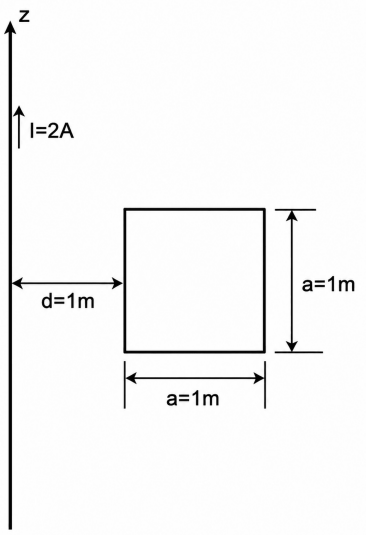


Fig. Q2.20.1

Q2.17.1 MCQ 2017 1M

A solid iron cylinder is placed in a region containing a uniform magnetic field such that the cylinder axis is parallel to the magnetic field direction. The magnetic field lines inside the cylinder will

A. bend closer to the cylinder axis

B. bend farther away from the axis

C. remain uniform as before

D. cease to exist inside the cylinder

Q2.17.2 NAT 2017 1M

The magnitude of magnetic flux density (B) in microteslas (μT), at the center of a loop of wire wound as a regular hexagon of side length 1 m, carrying a current $(I = 1$ A), and placed in a vacuum as shown in the figure is _____. (Give the answer up to two decimal places.)

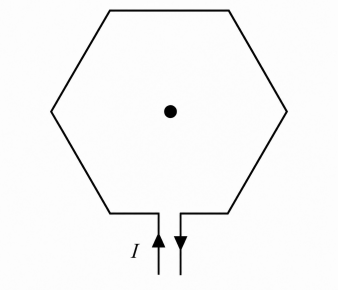


Fig. Q2.17.2

Q2.16.1 NAT 2016 2M

A rotating conductor of 1 m length is placed in a radially outward (about the z-axis) magnetic flux density (B) of 1 Tesla as shown in figure below. Conductor is parallel to and at 1 m distance from the z-axis. The speed of the conductor in r.p.m. required to induce a voltage of 1 V across it, should be _____.

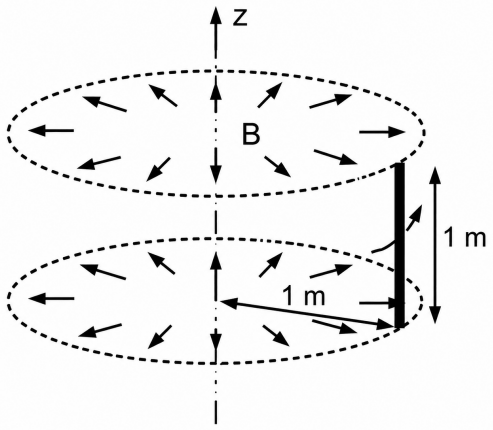


Fig. Q2.16.1

Q2.16.2 NAT 2016 1M

A soft-iron toroid is concentric with a long straight conductor carrying a direct current I. If the relative permeability μ_r of soft-iron is 100, the ratio of the magnetic flux densities at two adjacent points located just inside and just outside the toroid, is _____.

Q2.15.1 MCQ 2015 1M

A steady current I is flowing in the $-x$ direction through each of two infinitely long wires at $y = \pm\frac{L}{2}$ as shown in the figure. The permeability of the medium is μ_0. The $\vec{B}$-field at $(0, L, 0)$ is

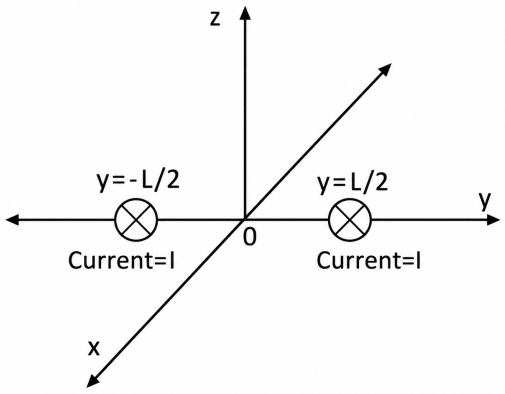


Fig. Q2.15.1

A. $-\frac{4\mu_0 I}{3\pi L}\hat{z}$ **B.** $+\frac{4\mu_0 I}{3\pi L}\hat{z}$ **C.** 0 **D.** $-\frac{3\mu_0 I}{4\pi L}\hat{z}$

Q2.15.2 NAT 2015 1M

Consider a one-turn rectangular loop of wire placed in a uniform magnetic field as shown in the figure. The plane of the loop is perpendicular to the field lines. The resistance of the loop is 0.4 Ω, and its inductance is negligible. The magnetic flux density (in Tesla) is a function of time, and is given by

$$B(t) = 0.25 \sin \omega t,$$

where $\omega = 2\pi \times 50$ radian/second. The power absorbed (in Watt) by the loop from the magnetic field is _____.

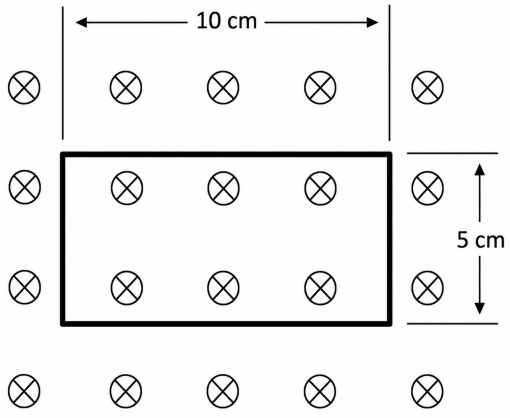


Fig. Q2.15.2

Q2.15.3 NAT 2015 1M

A circular turn of radius 1 m revolves at 60 rpm about its diameter aligned with the x-axis as shown in the figure. The value of μ_0 is $4\pi \times 10^{-7}$ in SI unit. If a uniform magnetic field intensity $\vec{H} = 10^7\,\hat{z}$ A/m is applied, then the peak value of the induced voltage, V_{turn} (in Volts), is _____.

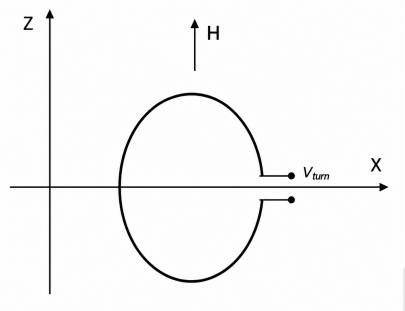


Fig. Q2.15.3

Q2.14.1 MCQ 2014 2M

The magnitude of magnetic flux density ($\vec{B}$) at a point having normal distance d meters from an infinitely extended wire carrying current of I A is $\frac{\mu_0 I}{2\pi d}$ (in SI units). An infinitely extended wire is laid along the x-axis and is carrying current of 4 A in the $+ve$ x direction. Another infinitely extended wire is laid along the y-axis and is carrying 2 A current in the $+ve$ y direction. μ_0 is permeability of free space. Assume $\hat{i}$, $\hat{j}$, $\hat{k}$ to be unit vectors along x, y and z axes respectively.

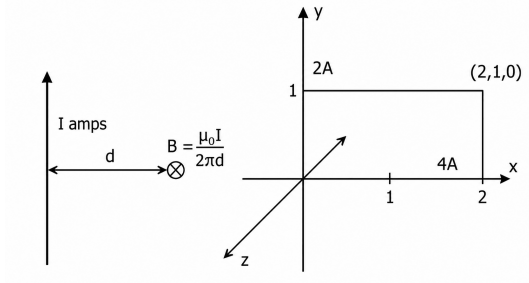


Fig. Q2.14.1

Assuming right handed coordinate system, magnetic field intensity, $\vec{H}$ at coordinate $(2, 1, 0)$ will be

A. $\frac{3}{2\pi}\hat{k}$ weber/m^2 **B.** $\frac{4}{3\pi}\hat{i}$ A/m **C.** $\frac{3}{2\pi}\hat{k}$ A/m **D.** 0 A/m

Q2.14.2 MCQ 2014 2M

The following four vector fields are given in Cartesian coordinate system. The vector field which does not satisfy the property of magnetic flux density is

A. $y^2a_x + z^2a_y + x^2a_z$ **B.** $z^2a_x + x^2a_y + y^2a_z$
C. $x^2a_x + y^2a_y + z^2a_z$ **D.** $y^2z^2a_x + x^2z^2a_y + x^2y^2a_z$

Q2.14.3 NAT 2014 1M

Two identical coupled inductors are connected in series. The measured inductances for the two possible series connections are 380 μH and 240 μH. Their mutual inductance in μH is _____.

Q2.13.1 MCQ 2013 1M

The flux density at a point in space is given by $\mathbf{B} = 4x\,\hat{a}_x + 2ky\,\hat{a}_y + 8\,\hat{a}_z$ Wb/m^2. The value of the constant k must be equal to

A. -2 **B.** -0.5 **C.** $+0.5$ **D.** $+2$

Q2.08.1 MCQ 2008 2M

A coil of 300 turns is wound on a non-magnetic core having a mean circumference of 300 mm and a cross-sectional area of 300 mm^2. The inductance of the coil corresponding to a magnetizing current of 3 A will be
(Given that $\mu_0 = 4\pi \times 10^{-7}$ H/m)

A. 37.68 μH **B.** 113.04 μH
C. 37.68 μH **D.** 113.04 μH

Q2.06.1 MCQ 2006 2M

Consider the following statements with reference to the equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$$

1. This is a point format the continuity equation
2. Divergence of current density is equal to the decrease of charge per unit volume per unit at every point
3. This is Maxwell's divergence equation
4. This represents the conservation of charge

Select the correct answer

A. Only 2 and 4 are true
B. 1, 2 and 3 are true
C. 2, 3 and 4 are true
D. 1, 2 and 4 are true

Q2.06.2 MCQ 2006 1M

Which of the following statements holds for the divergence of electric and magnetic flux densities

A. Both are zero

B. These are zero for static densities but non zero for time varying densities

C. It is zero for the electric flux density

D. It is zero for the magnetic flux density

Q2.04.1 MCQ 2004 1M

The inductance of a long solenoid of length 1000 mm wound uniformly with 3000 turns on a cylindrical paper tube of 60 mm diameter is

A. 3.2 μH **B.** 3.2 mH **C.** 32.0 mH **D.** 3.2 H

Q2.03.1 MCQ 2003 1M

Two infinite strips of width w m in x direction as shown in Figure, are carrying forward and return currents of $+I$ and $-I$ in the z direction. The strips are separated by a distance of x m. The inductance per unit length of the configuration is measured to be L H/m. If the distance of separation between the strips is now reduced to $\frac{x}{2}$ m, the inductance per unit length of the configuration

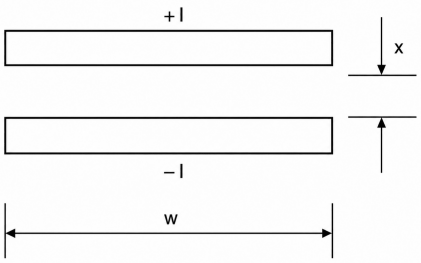


Fig. Q2.03.1

A. $2L$ H/m **B.** $\frac{L}{4}$ H/m **C.** $\frac{L}{2}$ H/m **D.** $4L$ H/m

Q2.03.2 MCQ 2003 1M

Two conductors are carrying forward and return current of $+I$ and $-I$ as shown in Figure. The magnetic field intensity $\vec{H}$ at point P is

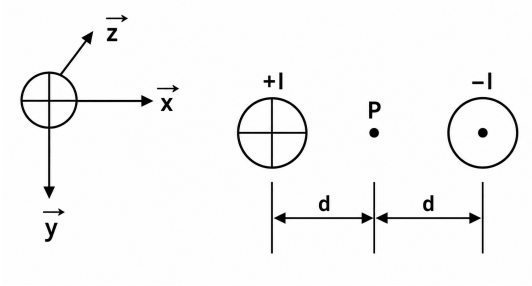


Fig. Q2.03.2

A. $\frac{I}{\pi d}\bar{y}$ **B.** $\frac{I}{\pi d}\bar{x}$ **C.** $\frac{I}{2\pi d}\bar{y}$ **D.** $\frac{I}{2\pi d}\bar{x}$

Q2.02.1 NAT 2002 5M

The magnetic vector potential in a region is defined by

$$\vec{A} = e^{-y}\sin(x)\,\hat{a}_z.$$

An infinitely long conductor, having a cross section area, $a = 5\,\text{mm}^2$ and carrying a dc current, $I = 5$ A in they direction, passes through this region as shown in Fig. P3.

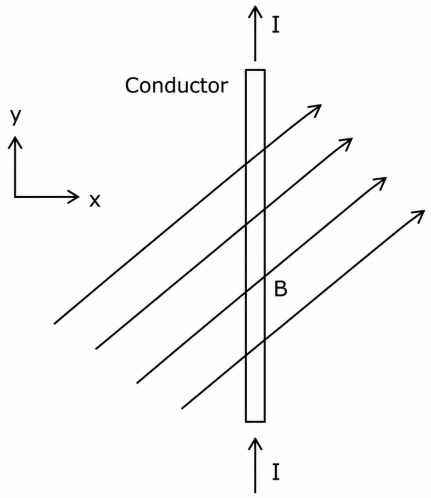


Fig. Q2.02.1

Determine the expression for (a) $\vec{B}$ and (b) force density $\vec{f}$ exerted on the conductor.

Q2.00.1 NAT 2000 5M

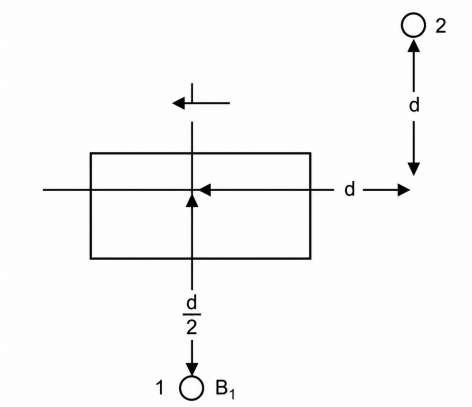


Fig. Q2.00.1

(a) Show via the construction of a suitable Gaussian surface, that the capacitance of a spherical capacitor consisting of two concentric shells of radii a and b is given by

$$C = 4\pi\varepsilon_0 \frac{ab}{(b-a)}$$

Where ε_0 is the free space permittivity.

(b) A current I in the short conducting element shown in fig produces a flux density B_1 at point 1.

Determine the magnitude and the direction of the flux density vector at point 2.

Q2.00.2 MCQ 2000 1M

An electron with velocity u is placed an electric field E and magnetic field B, the force experienced by the electron is given by

A. $-eE$ **B.** $-eu \times B$

C. $-e\,(u \times E + B)$ **D.** $-e\,(E + u \times B)$

Q2.99.1 MCQ 1999 1M

Two parallel wires separated by a distance d are carrying a current I in the same direction. The magnetic field along a line running parallel to these wires and midway between them

A. Depends upon I

B. Is zero

C. Depends upon d

D. Depends upon the permeability of medium between the wires

Q2.99.2 MCQ 1999 1M

An electromagnetic field is radiated from

A. A stationary point charge

B. A capacitor with a DC voltage

C. A conductor carrying a DC current

D. An oscillating dipole

Q2.98.1 NAT 1998 5M

An electron moves in the X-Y plane with a speed of 10^6 m/s. Its velocity vector makes an angle of 60° with X-axis. A magnetic field of magnitude 10^{-2} T exists along the Y-axis. Compute the magnetic force exerted on the electron and its direction.

Q2.98.2 MCQ 1998 1M

The laws of electromagnetic induction (Faraday's and Lenz's law) are summarized in the following equation:

A. $e = iR$ **B.** $e = L\dfrac{di}{dt}$

C. $e = -\dfrac{d\Psi}{dt}$ **D.** None of the above

Q2.97.1 NAT 1997 5M

A square coil of turns and 10 cm side is moved through a steady magnetic field of 1 Wb/m^2 at a constant velocity of 2 m/sec with its plane perpendicular to the field as shown in figure. Plot the variation of induced e.m.f as the coil moves along the field.

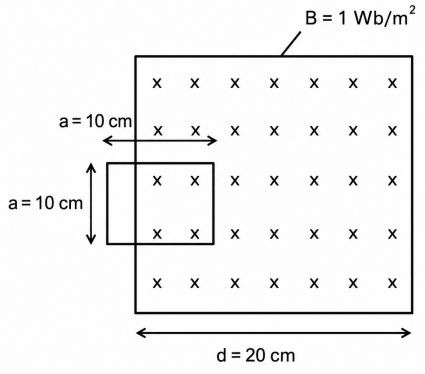


Fig. Q2.97.1

Q2.97.2 NAT 1997 5M

An infinitely long straight wire carries 1000 A of current and in the vicinity there is a circular conducting loop of 100 mm diameter with the center of the loop 1 m away from the straight conductor. Both the wire and the loop are coplanar. Determine the magnitude and direction of current in the loop that produces a zero flux density at its center.

Q2.96.1 MCQ 1996 1M

The energy stored in the magnetic field of solenoid 30 cm long and 3 cm diameter wound with 1000 turns of wire carrying a current of 10 A is

A. 0.015 joule **B.** 0.15 joule
C. 0.5 joule **D.** 1.15 joule

Q2.95.1 NAT 1995 5M

A steady current I flows through a long cylindrical wire of radius R, find the magnetic vector potential at any point outside the conductor at radius r. Vector potential on the surface of the conductor may be taken to be equal to zero.

Q2.95.2 MCQ 1995 1M

A monochromatic plane electromagnetic wave travels in vacuum in the positive X direction (x, y, z system of coordinates). The electric and magnetic fields can be expressed as

A. $E(x,t) = E_0 \cos(kx - \omega t)\, \bar{a}_y$
$H(x,t) = H_0 \cos(kx - \omega t)\, \bar{a}_z$

B. $E(x,t) = E_0 \cos(kx - \omega t)\, \bar{a}_y$
$H(x,t) = H_0 \cos\left(kx - \omega t - \frac{\pi}{2}\right) \bar{a}_z$

C. $E(x,t) = E_0 \cos(kx - \omega t)\, \bar{a}_y$
$H(x,t) = -H_0 \cos(kx - \omega t)\, \bar{a}_z$

D. $E(x,t) = E_0 \cos(kx - \omega t)\, \bar{a}_z$
$H(x,t) = H_0 \cos(kx - \omega t)\, \bar{a}_y$

Q2.94.1 NAT 1994 1M

The inductance of a coil is proportional to the _____ of the number of turns, all over parameters remaining the same.

Q2.94.2 MCQ 1994 1M

Match the appropriate item from the right hand side with those on the left hand side.

(a) Line charge	(P) Maxwell
(b) Magnetic flux density	(Q) Poynting's Vector
(c) Displacement current	(R) Transmission line conductors
(d) Power flow	(S) Biot-Savart's law
	(T) Gauss's law
	(U) Faraday's law

Q2.94.3 NAT 1994 1M

Static magnetic fields induce currents in closed conducting loops.

Q2.94.4 MCQ 1994 1M

When a transformer winding suffers a short-circuit, the adjoining turns of the same winding experience

A. an attractive force **B.** a repulsive force
C. no force **D.** none of the above

Q2.93.1 MCQ 1993 1M

The line integral of the vector potential A around the boundary of a surface S represents

A. Flux through in the surface S

B. Flux density in the surface S

C. Magnetic density

D. Current density

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q2.26.1 **B**	Q2.24.1 **2**	Q2.23.1 **B,D**
Q2.23.2 **2.49 to 2.51**	Q2.22.1 **C**	Q2.22.2 **B**
Q2.21.1 **A**	Q2.21.2 **100**	Q2.20.1 **138.33**
Q2.17.1 **A**	Q2.17.2 **0.6928**	Q2.16.1 **9.55**
Q2.16.2 **100**	Q2.15.1 **A**	Q2.15.2 **0.1927**
Q2.15.3 **248.05**	Q2.14.1 **C**	Q2.14.2 **C**

Q2.14.3 35	Q2.13.1 A	Q2.08.1 B
Q2.06.1 D	Q2.06.2 D	Q2.04.1 C
Q2.03.1 C	Q2.03.2 A	Q2.02.1 *
Q2.00.1 *	Q2.00.2 D	Q2.99.1 B
Q2.99.2 D	Q2.98.1 0.8×10^{-15}	Q2.98.2 C
Q2.97.1 *	Q2.97.2 15.92	Q2.96.1 B
Q2.95.1 *	Q2.95.2 A	Q2.94.1 2
Q2.94.2 a:R, b:S, c:P, d:Q	Q2.94.3 False	Q2.94.4 A
Q2.93.1 A		

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